Sliding Window — Lab Exam Algorithms

Sliding Window — Sender Algorithm

1. Start the sender program.

2. Create a `ServerSocket` on port `5000` and wait for the receiver to connect.

```
ServerSocket server = new ServerSocket(5000);
Socket socket = server.accept();
```

3. Create input and output streams for communication with the receiver.

```
BufferedReader in =
    new BufferedReader(
        new InputStreamReader(socket.getInputStream()));

PrintWriter out =
    new PrintWriter(
        socket.getOutputStream(), true);
```

4. Create a `Scanner` object to read input from the user.

```
Scanner sc = new Scanner(System.in);
```

5. Ask the user to enter the number of frames.

```
System.out.print("Enter the no. of frames: ");
n = Integer.parseInt(sc.nextLine());
```

6. Send the number of frames to the receiver.

```
out.println(n);
```

7. Check whether the number of frames is within the window size.

```
if (n <= 7)
```

8. Ask the user to enter the messages for the frames.

```
System.out.println("Enter " + n + " Messages to be sent");
```

9. Read each frame message and send it to the receiver.

```
for (int i = 0; i < n; i++)
{
    String msg = sc.nextLine();
    out.println(msg);
}
```

10. Receive the acknowledgment from the receiver.

```
ack = Integer.parseInt(in.readLine());
```

11. Display the acknowledgment received.

```
System.out.println("Acknowledgment received for " + ack + " frames");
```

12. If the number of frames exceeds the window size, display an error message and stop.

```
else
{
    System.out.println("The no. of frames exceeds window size");
    break;
}
```

13. Ask the user whether more frames should be sent.

```
System.out.print("Do you want to send some more frames: ");
choice = sc.nextLine();
```

14. Send the choice to the receiver and repeat if the choice is `"yes"`.

```
out.println(choice);

} while (choice.equals("yes"));
```

15. Close the socket and server.

```
socket.close();
server.close();
```

16. Stop the program.

Sliding Window — Receiver Algorithm

1. Start the receiver program.

2. Create a socket and connect to the sender using `localhost` and port `5000`.

```
Socket socket =
    new Socket("localhost", 5000);
```

3. Create input and output streams for communication with the sender.

```
BufferedReader in =
    new BufferedReader(
        new InputStreamReader(socket.getInputStream()));

PrintWriter out =
    new PrintWriter(
        socket.getOutputStream(), true);
```

4. Initialize the frame pointer to `-1`.

```
int pointer = -1;
```

5. Receive the number of frames from the sender.

```
n = Integer.parseInt(in.readLine());
```

6. Check whether the number of frames is within the window size.

```
if (n <= 7)
```

7. Receive each frame from the sender.

```
for (int i = 0; i < n; i++)
{
    pointer = ++pointer % 8;
    String msg = in.readLine();
```

8. Display the received frame number and its message.

```
System.out.println("The received Frame " + pointer + " is: " + msg);
}
```

9. Display that the acknowledgment has been sent.

```
System.out.println("\nAcknowledgment sent");
```

10. Send the acknowledgment to the sender.

```
out.println(pointer + 1);
```

11. If the number of frames exceeds the window size, terminate the loop.

```
else
{
    break;
}
```

12. Receive the sender's choice about sending more frames.

```
choice = in.readLine();
```

13. Repeat the process if the choice is `"yes"`.

```
} while (choice.equals("yes"));
```

14. Close the socket.

```
socket.close();
```

15. Stop the program.